Exercise 3 (CASE statements)

1.
```
SELECT    product_name,
          price
CASE
    WHEN  price >1000 THEN 'Expensive'
    WHEN  price BETWEEN 100 AND 1000 THEN 'Mid-range'
    ELSE 'Budget'
END AS price_category
FROM products;
```

product_name	price	price_category
Laptop	1200.00	Expensive
Phone	800.00	Mid-range
keyboard	45.00	Budget
Monitor	300.00	~~Budget~~ Mid-range
Mouse	25.00	Budget

2.
```
SELECT    customer_name,
          amount
CASE
    WHEN  amount >= 1000.00 THEN 'High Value'
    WHEN  amount BETWEEN 500.00 AND 999.99 THEN 'Medium Value'
    ELSE 'Low Value'
END AS order_value_category
FROM orders;
```

customer-name	amount	order_value_category
Alice	150.00	Low Value
Bob	560.00	Medium value
Charlie	999.99	Medium Value
Diana	45.50	Low Value
Ethan	1200.00	High Value

3. SELECT emp-name,
department,
salary

CASE
WHEN department = 'IT' AND salary > 80000 THEN 'Senior IT'
WHEN department = 'HR' AND salary > 55000 THEN 'Experienced
ELSE 'Staff'
END AS position-level
FROM employees;

emp-name	department	salary	position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

4. SELECT student_name,
score

CASE
WHEN score >= 90 THEN 'A'
WHEN score BETWEEN 80 AND 89 THEN 'B'
WHEN score BETWEEN 70 AND 79 THEN 'C'
WHEN score BETWEEN 60 AND 69 THEN 'D'
ELSE 'F'
END AS grade
FROM students;

student_name	score	grade
Anna	92	'A'
Ben	76	'C'
Cara	59	'F'
David	83	'B'
Ella	68	'D'

5. SELECT delivery_id,
delivery_time_minutes
CASE
WHEN delivery_time_minutes <= 30 THEN 'Fast'
WHEN delivery_time_minutes BETWEEN 31 AND 60 THEN 'On Time'
ELSE 'Late'
END AS performance
FROM deliveries;

delivery_id	delivery_time_minutes	perfomance
1	45	On Time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

6.
```
SELECT issue_type,
       priority
CASE
  WHEN priority =3 THEN 'High'
  WHEN priority =2 THEN 'Medium'
  ELSE 'Low'
END AS priority_label
FROM tickets;
```

issue_type	priority	priority_label
Login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

7. SELECT student_id,
(days_present * 100 / total_days) AS attendance_percentage
CASE
WHEN days_present*100/total_days >= 90 THEN 'Excellent'
WHEN days_present*100/total_days BETWEEN 75 AND 89
THEN 'Good'
ELSE 'Needs Improvement'
END AS attendance_status
FROM attendance;

student_id	attendance %	attendance status
1	90	Excellent
2	60	Needs improvement
3	96	Excellent
4	50	Needs improvement
5	100	Excellent

8. SELECT product_id,
stock_qty
CASE
WHEN stock_qty = 0 THEN 'Out of stock'
WHEN stock_qty >= 1 AND stock_qty < 5 THEN
'Low stock'
ELSE 'In stock'
END AS stock_status
FROM products_inventory;

product_id	stock_qty	stock_status
1	5	In stock
2	0	Out of stock
3	25	In stock
4	10	In stock
5	3	Low stock

9.
```
SELECT    subject,
          enrolled_students
CASE
WHEN      enrolled_students >= 25   THEN   'Large'
WHEN      enrolled_students BETWEEN 10 AND 24  THEN 'Medi
  ELSE    'Small'
END AS    class_size_category
FROM      classes;
```

Subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

10.
```
SELECT    payment_id,
          payment_method,
          amount
CASE
  WHEN    payment_method = 'Cash' AND amount >= 200
    THEN  'Eligible for Discount'
    ELSE  'Not Eligible'
END AS    discount_eligibility
FROM      payments;
```

payment_id	payment_method	amount	discount_eligibility
1	Card	50.00	Not eligible
2	Cash	200.00	Eligible for discount
3	Card	150.00	Not eligible
4	Paypal	75.00	Not eligible
5	Cash	300.00	Eligible for discount